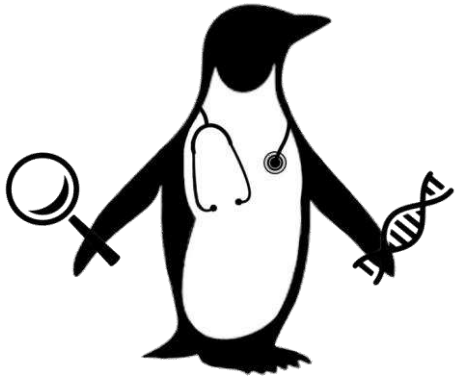


PENGQUIN

PENGQUIN:

PErsoNal Genome and QUery in healthcare and clinical practice

PhD Candidate: Elias Crum

VITO 18-month jury
Nov 20th, 2025

Advisors: Dr. Gökhan Ertaylan[1], Dr. Bart Buelens[1], Dr. Ruben Taelman[2], Prof. Dr. Ruben Verborgh[2]

1 = vito 2 = GHENT UNIVERSITY

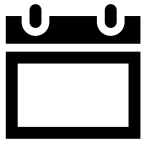
Presentation Overview

Project Review

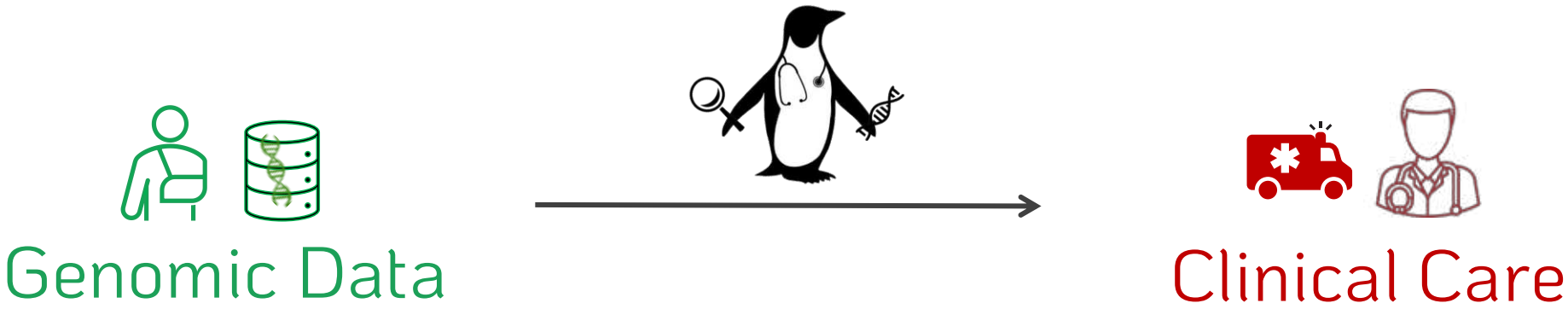
~24 Month Progress

Current Undertakings

Remaining PhD Plans

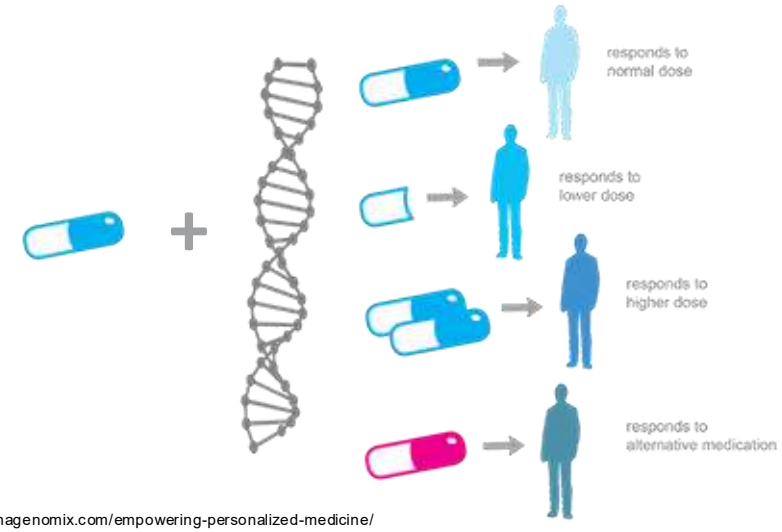


PENGQUIN: PErsoNal Genome QUery IN healthcare and clinical practice



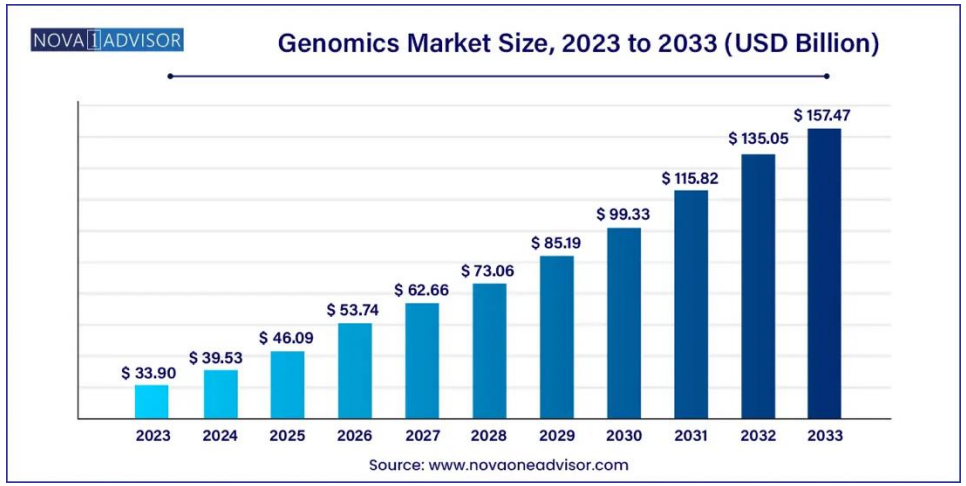
Genomics + Healthcare

A *Example: Pharmacogenomics*



https://alphagenomix.com/empowering-personalized-medicine/

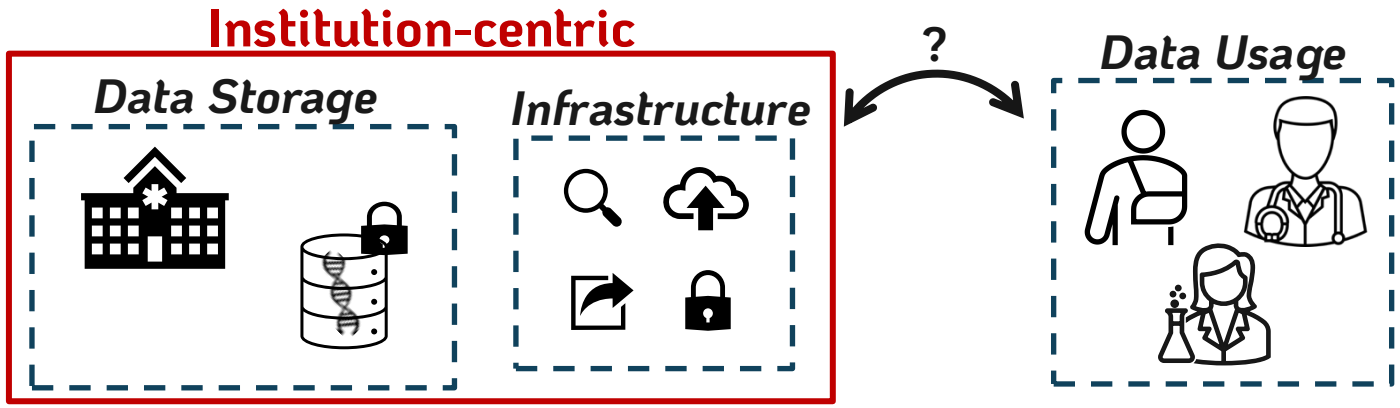
B *Genomics market is growing RAPIDLY*



https://www.novaoneadvisor.com/report/genomics-market

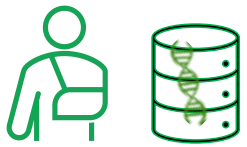
C

Problem:

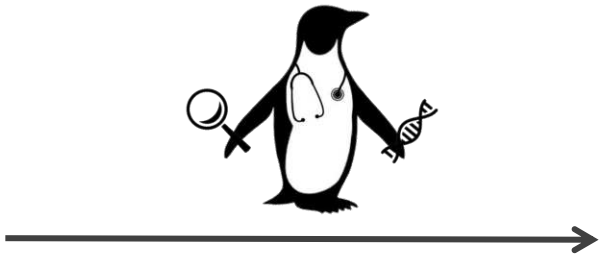


➔ ***Low Efficiency + High Costs***

PENGQUIN: PErsoNal Genome QUery IN healthcare and clinical practice



Genomic Data



Clinical Care

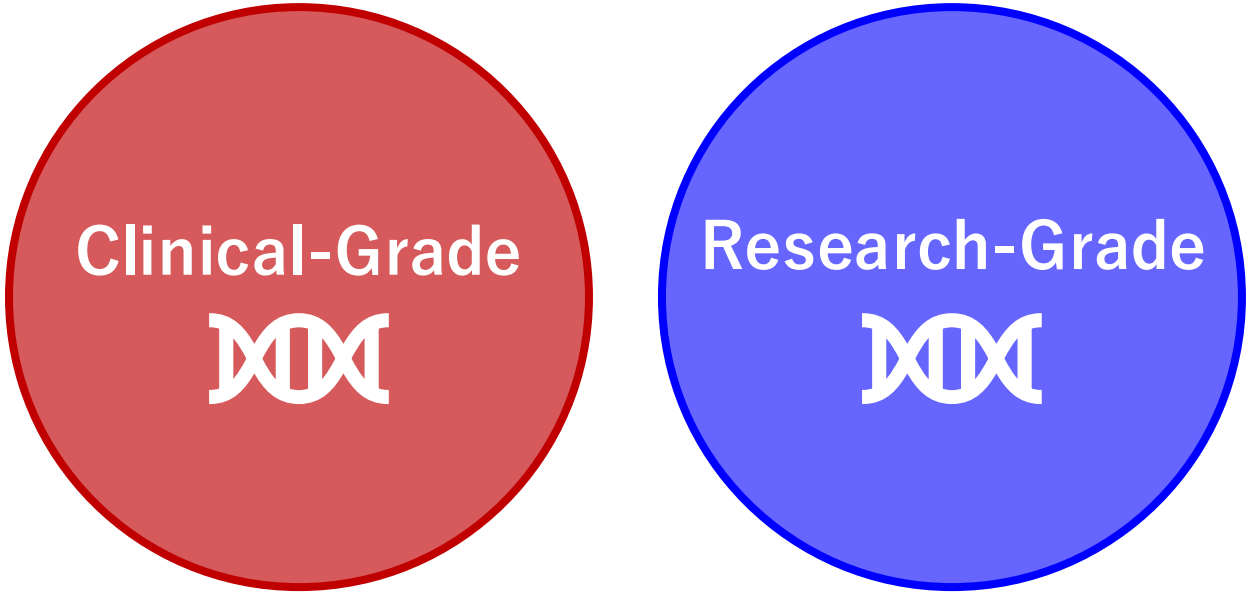
Private, Semantic Linked Data

Query Capabilities

Framework + App

Genomic Data for Clinical Use

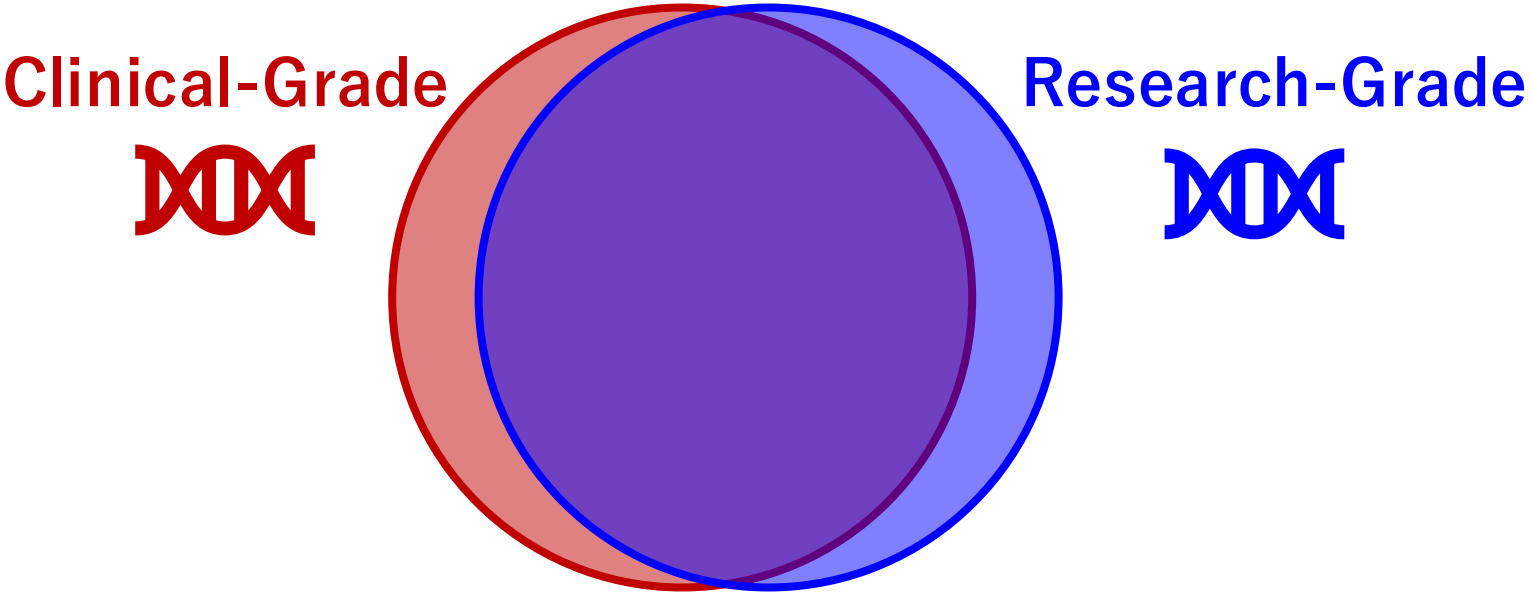
- Format is use-case specific
- Whole Genome Sequence (WGS) data
- Large Projects: 1+ Million Genomes (EU) [1]



[1] https://digital-strategy.ec.europa.eu/en/policies/1-million-genomes

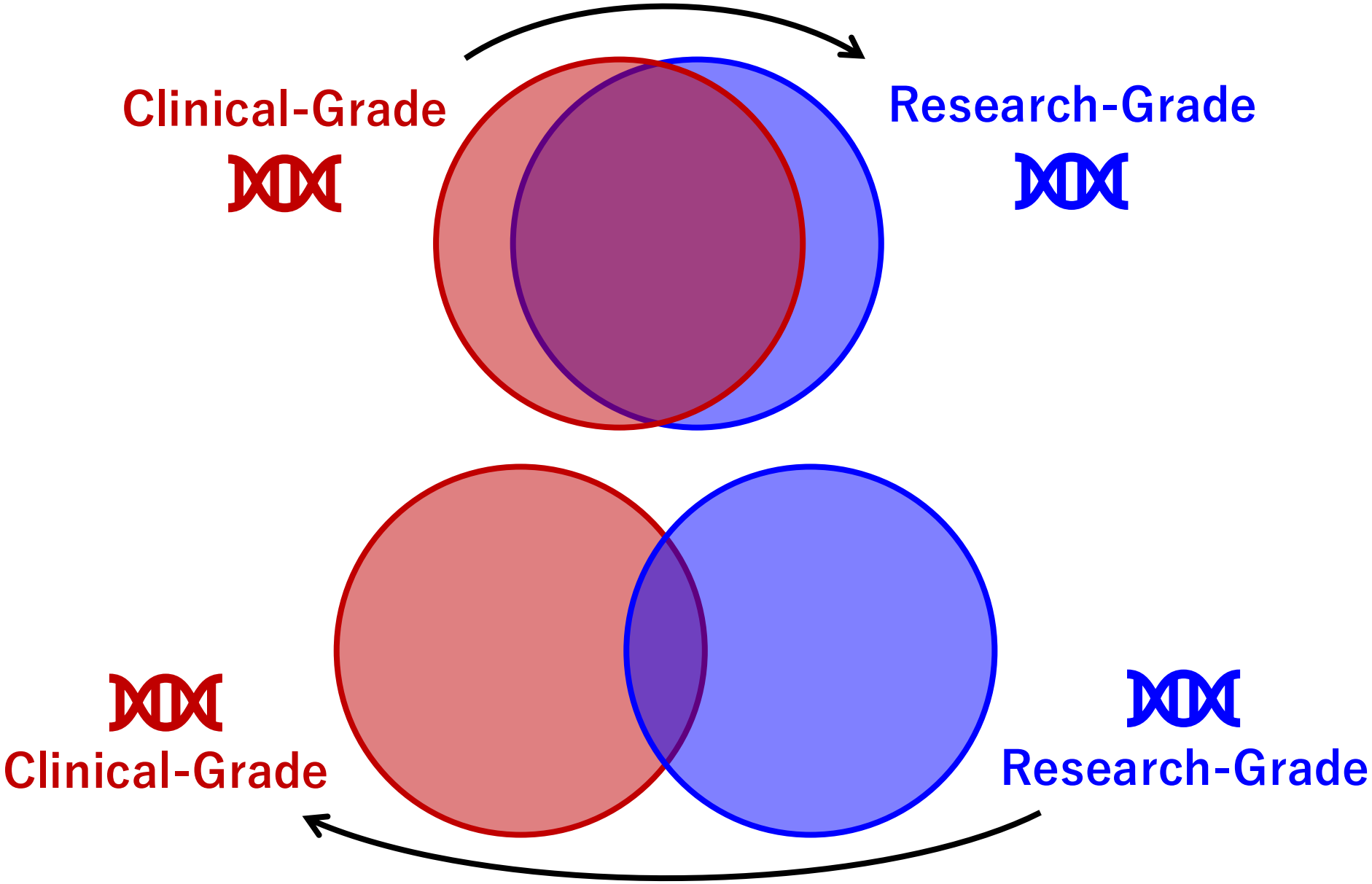
Genomic Data for Clinical Use (assumption)

- Format is use-case specific
- Whole Genome Sequence (WGS) data
- Large Projects: 1+ Million Genomes (EU) [1]



[1] https://digital-strategy.ec.europa.eu/en/policies/1-million-genomes

Genomic Data for Clinical Use (reality)



The current landscape

Intended use is VERY important

Clinical use ≠ Research use

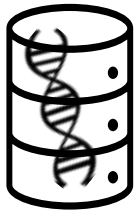
** Funding goes predominantly to RESEARCH-GRADE **

- Cheaper
- Fewer regulatory challenges
- Funders / Policy makers misunderstanding

** But how do we “translate” to clinical use?? **

Sharing Genomic Data – Technology Considerations

Key Implementation Factors



Data Storage

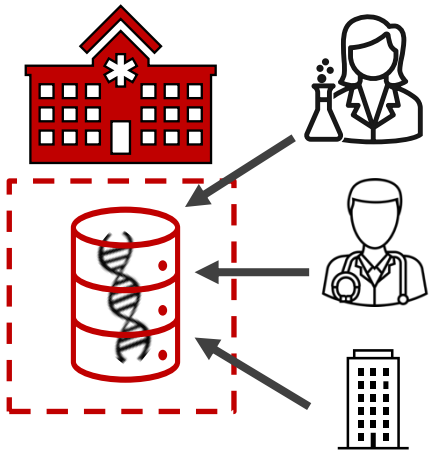
Considerations:

- patient consent
- data security
- access controls

&

Considerations:

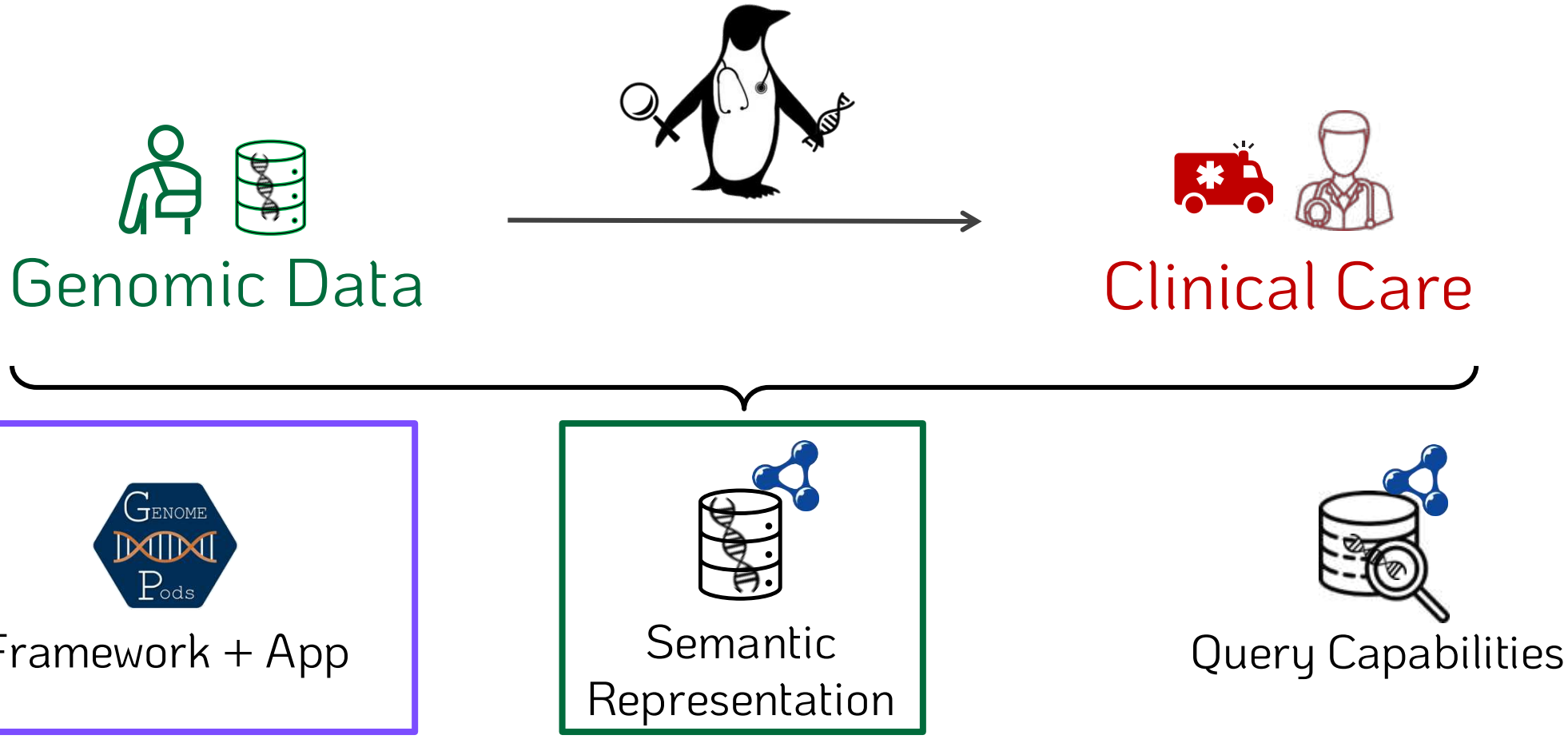
- consent changes
- governance
- infrastructure



Data Sharing

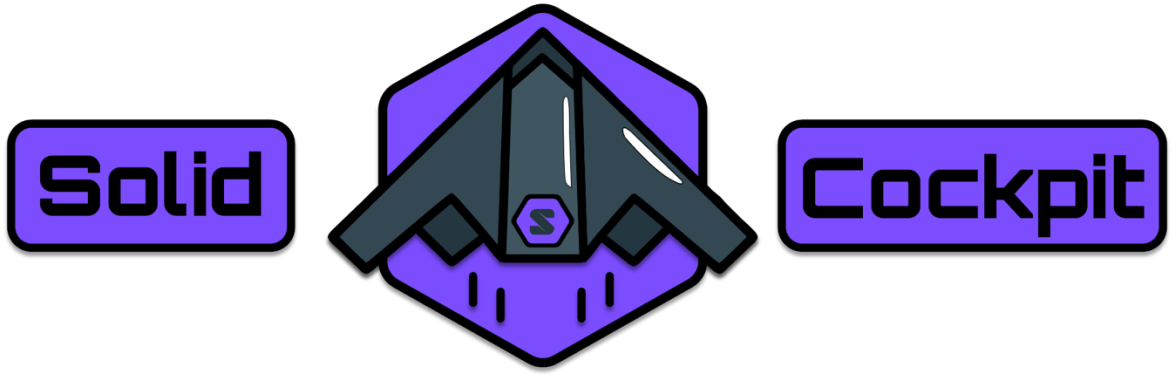
A more in-depth analysis of these challenges + potential solutions detailed in an upcoming publication in *Computers in Biology and Medicine*

PENGQUIN: PErsoNal Genome QUery IN healthcare and clinical practice



1. Data Interaction Framework

A Web Application to:



- *Upload Data to a Solid Pod*
- *Browse contents of a Solid Pod*
- *Edit Privacy of (and share) data in a Solid Pod*
- *Query Data in Solid Pods + SPARQL Endpoints*
  - *Includes a query cache for storing queries and results*

*** Enables patient, provider, and auditor control over data ***

What is a Solid Pod?

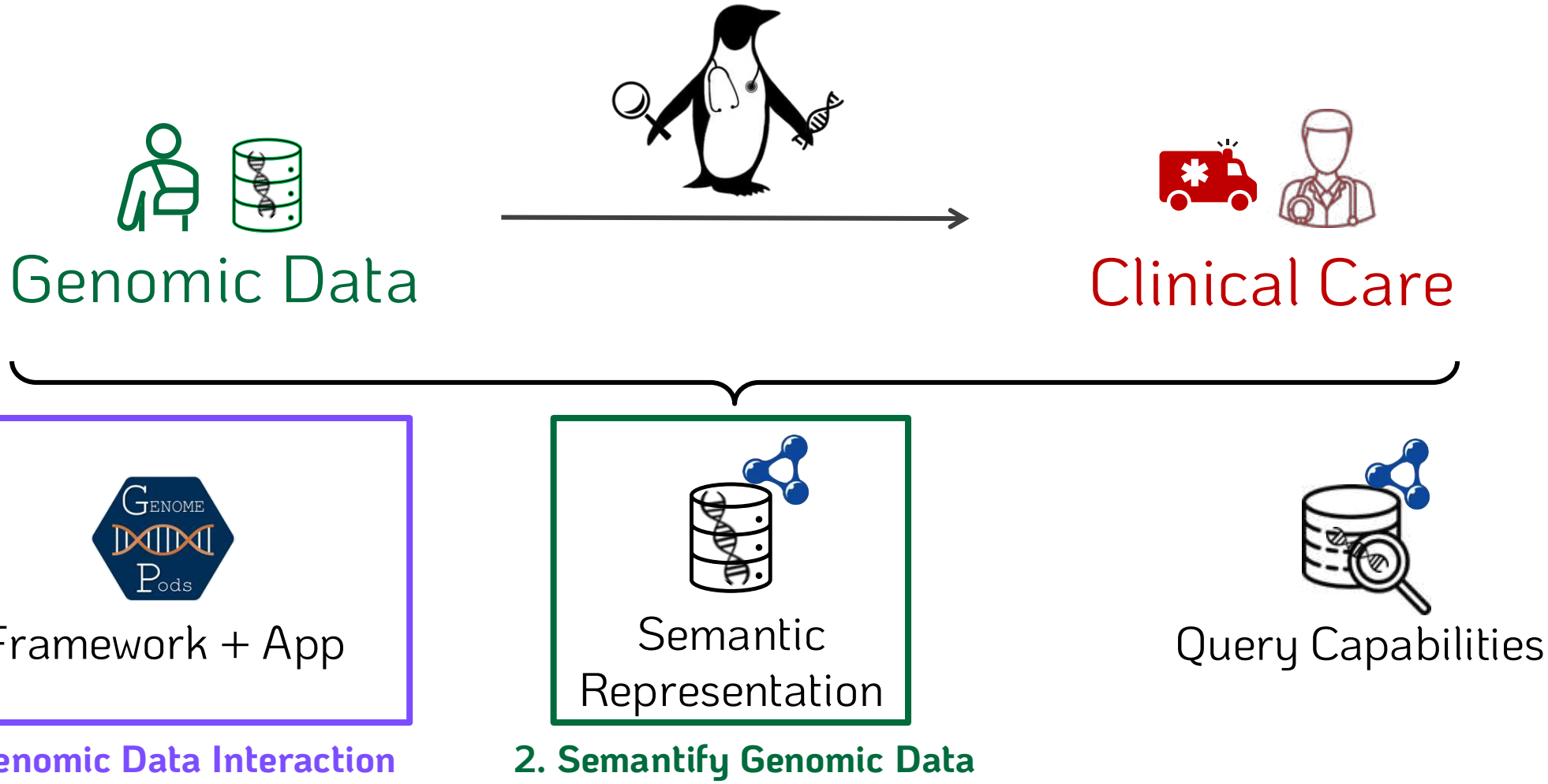
Secure, Decentralized Data Storage Specification

- *Pod exists on a Server (accessible to the Web)*
- *Can store files, RDF data, and Linked Data*
- *Provides control policies for stored data*

*** Enables secure, authenticated sharing via HTTP protocols ***

https://solidproject.org/about

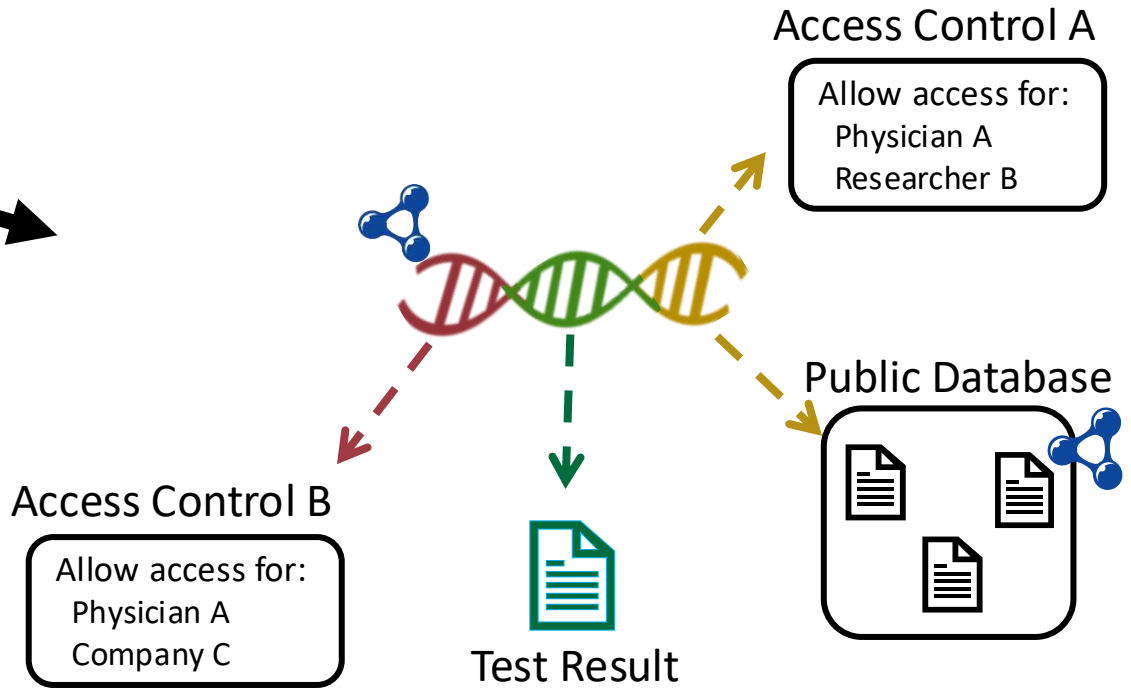
PENGQUIN: PErsoNal Genome QUery IN healthcare and clinical practice



2. Genomic Data Representation

From a file-based approach

To a granular, dynamic RDF approach



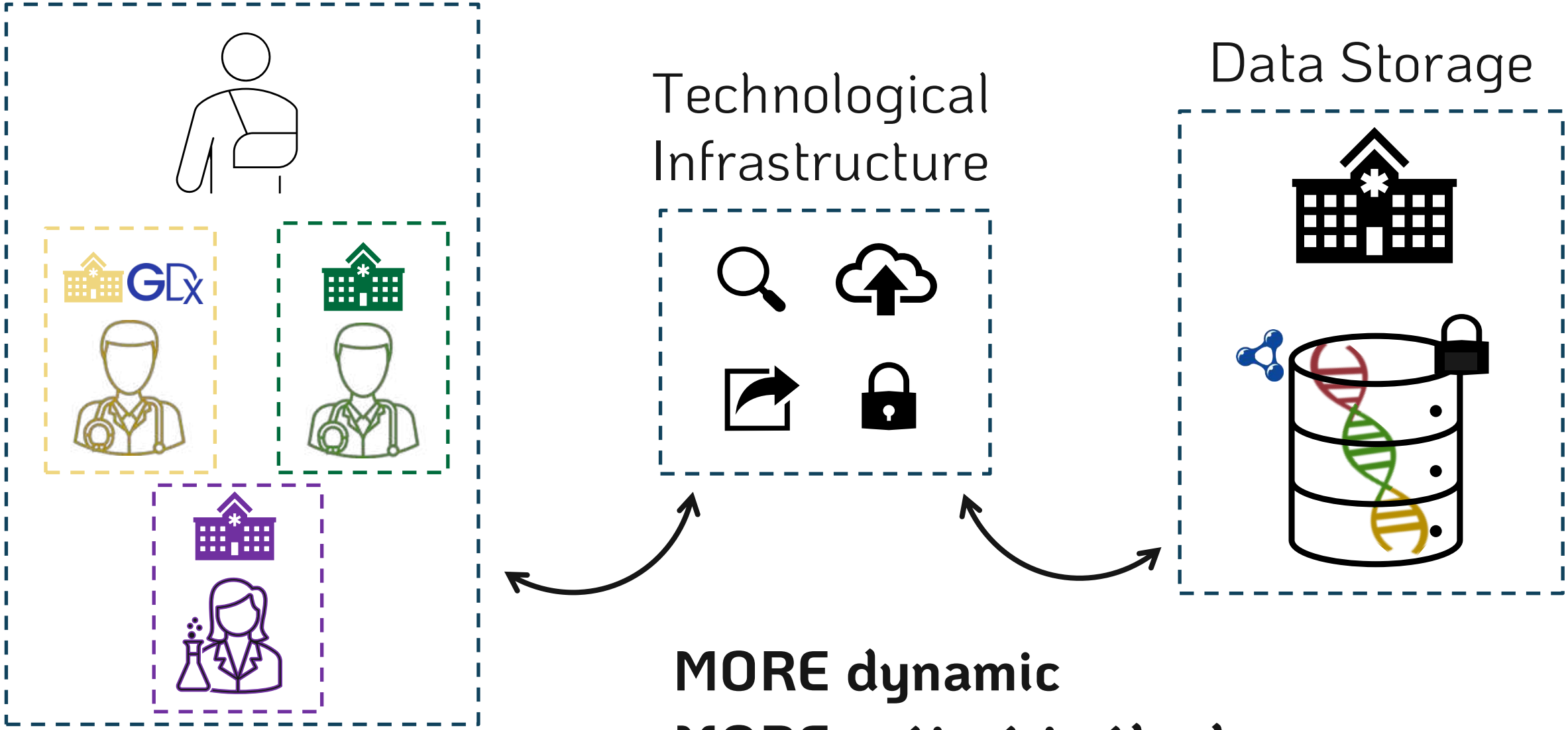
Conversion using existing ontologies

Policy definition via ODRL[1]

Implementation within Solid Pod Env

[1] https://www.w3.org/TR/odrl-model/

Future Possibilities?



MORE dynamic

MORE patient in-the-loop

Thank you!

More PhD specifics

LinkedIn

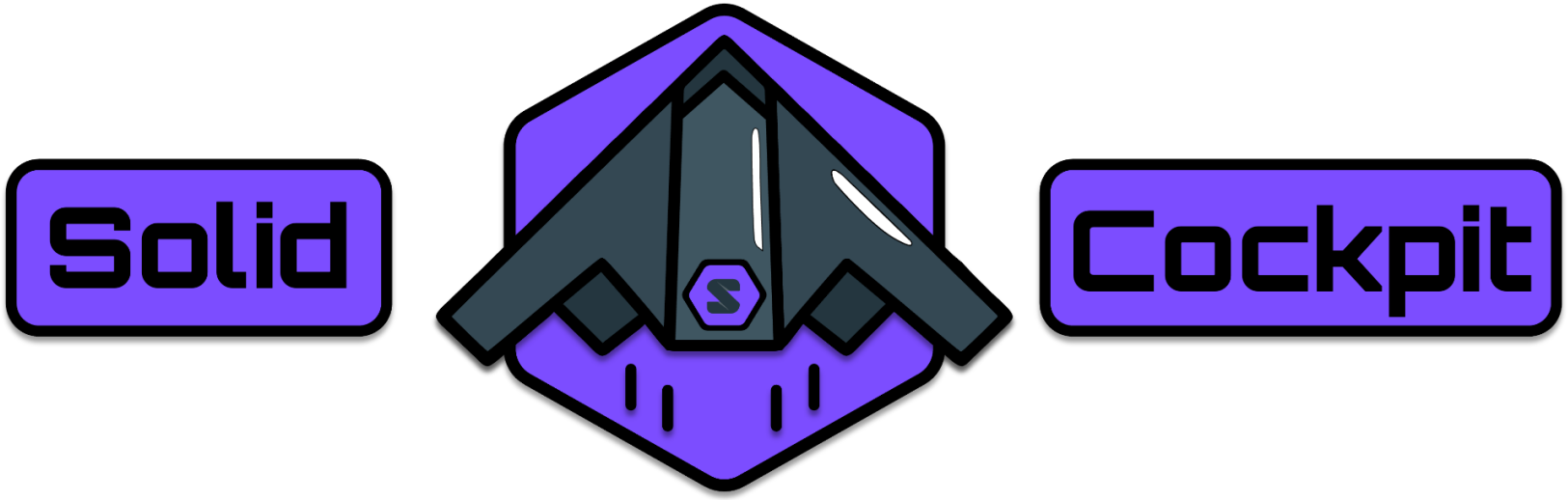
GitHub

Advisors: Dr. Gökhan Ertaylan[1], Dr. Bart Buelens[1], Dr. Ruben Taelman[2], Prof. Dr. Ruben Verborgh[2]

1 = vito 2 = GHENT UNIVERSITY

Informal Demo / Tutorial Later Today

(for those interested)



knowledgeonwebscale.github.io/solid-cockpit/

Supplemental Slides

Genomic Data

Clinical-Grade

Higher read quality

More expensive

Stricter use-regulations

Research-Grade

Lower read quality

Less expensive

Flexible use-regulations

Dimension	Clinical Grade Genome Data Sharing	Research Grade Genome Data Sharing
1. Primary Purpose and Scope	Patient care and medical decision-making.	Discovery and hypothesis-driven studies.
2. Data Accuracy and Quality	Stringent quality control (QC) and validation.	High but no absolute QC standards.
3. Regulatory Requirements	Subject to medical regulatory frameworks.	Subject to research ethics regulations.
4. Consent Requirements	Clinical-informed consent.	Research-informed consent.
5. Privacy and Confidentiality	Highest level of patient privacy required.	Strong but variable privacy measures.
6. Security Controls	Robust, enterprise-grade security.	Secure but more flexible.
7. Data Governance and Oversight	Hospital/clinic-based governance.	Institution or consortium-based governance.
8. Ethical Considerations	Focus on direct patient benefit and “do no harm.”	Focus on scientific advancement and participant welfare.
9. Infrastructure Requirements	Highly reliable, clinically validated systems.	Flexible, scalable research infrastructure.
10. Architecture Requirements	Standardized, interoperable clinical systems.	Flexible, modular architectures.
11. Legal Liabilities	High.	Moderate.
12. Interoperability	High necessity for clinical integration.	Variable.
13. Data Ownership and Control	Patient-oriented ownership with institutional custodianship.	Institution or sponsor-owned.
14. Data Sharing Scope	Limited to necessary clinical stakeholders.	Broad or consortium-based.
15. Cost and Funding Models	Healthcare reimbursement or institutional budgets.	Grant-funded or institutional research budgets.

Genomic Sequence Data Sharing for Clinical Practice: A Scoping Review

vito

GHENT UNIVERSITY

Aim: Identify existing implementations, guidelines, and challenges of clinical sharing of genomic data.

Methods: Reviewed 55 studies, 57 national genomics projects, and 20 major genomics companies.

Results: Clinical genome sharing implementations exist, yet scalability and widespread adoption remain limited.

Suggested steps forward:

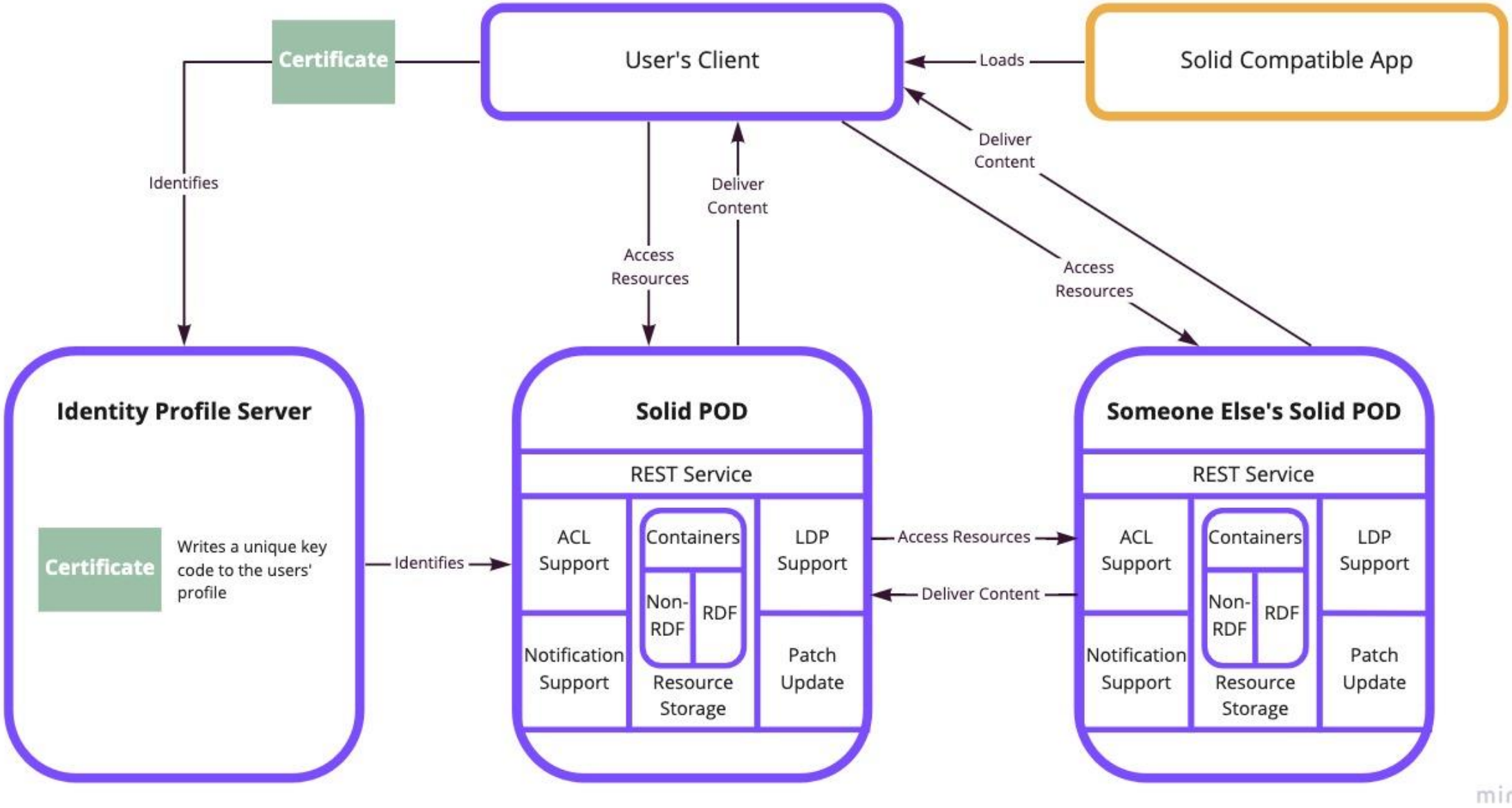
Harmonize legal bases for secondary clinical use of research data

Mandate (GA4GH) technical standards

Scale laboratory accreditation & continual pipeline validation

Align economic incentives

Solid's Architecture



https://solid.georgetown.domains/technical-component/

Solid Pod Implementations?

BBC MyPDS App

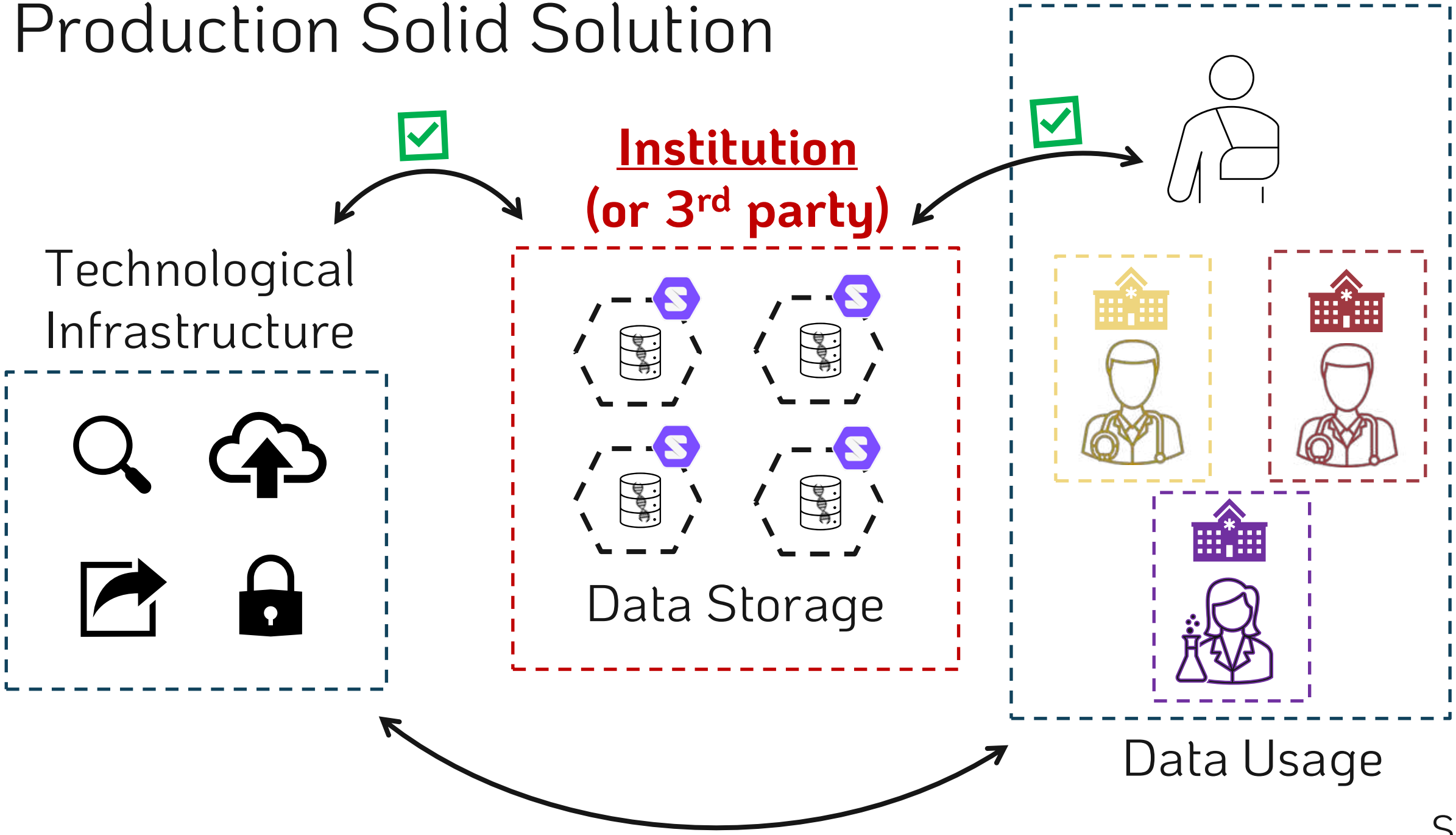
Flemish Data Utility Company

health data vaults

Flemish Smart Dataspace

Solid Lab Use-cases

A Production Solid Solution



Semantic VCF – *Sequencing Data Formats Intro*

FASTQ ✚ **FASTA** → **SAM/BAM** → **VCF**

Raw Sequencing Outputs

Contiguous DNA sequence

FASTQ reads aligned to a FASTA reference

Areas in SAM where sample and reference differ

```
1  >gi|568336023|gb|CM000663.2| Homo sapiens chromosome 1, GRCh
2  CCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAACCCTAAC
3  ACCCTAACCCTAACCCTAACCCTAACCCAACCCTAACCCTAACCCTAACCCTAACCCTAA
4  TAACCCTAACCCTAACCCTAACCCTAACCTAACCCTAACCCTAACCCTAACCCTAACCCT
5  CTAACCCTAACCCCTAACCCTAACCCTAAACCCTAAACCCTA
6  CAACCCCAACCCCAACCCCAACCCCAACCCCAACCCTAACCC
7  CCTAACCCTAACCCTAACCCTAACCCTAACCCCTAACCCCTA
8  TAACCCTAACCCCTAACCCTAACCCTAACCCTAACCCTCGCG
9  TCTGACCTGAGGAGAACTGTGCTCCGCCTTCAGAGTACCACC
10 CGCCCTCGCGGTGCTCTCCGGGTCTGTGCTGAGGAGAACGCA
11 CGGCGCAGGCGCAGAGAGGCGCGCCGCGCCGGCGCAGGCGCA
12 AGAGAGGCGCGCCGCGCCGGCGCAGGCGCAGAGAGGCGCGCC
13 CGCGCCGGCGCAGGCGCAGACACATGCTAGCGCGTCGGGGTG
14 GCCGCGCCGGCGCAGGCGCAGAGACACATGCTACCGCGTCCA
15 AGGCGCACCGCGCCGGCGCAGGCGCAGAGACACATGCTAGCG
16 GCAGAGACGCAAGCCTACGGGCGGGGGTTGGGGGGGCGTGTG
17 GCTGGGGCGGGGGGAGGGTGGCGCCGTGCACGCGCAGAAACT
18 GGTAGAACCTCAGTAATCCGAAAAGCCGGGATCGACCGCCCC
19 GCTTGCTCACGGTGCTGTGCCAGGGCGCCCCCTGCTGGCGAC
20 GAGTGGTGGCCAGCGCCCCCTGCTGGCGCCGGGGCACTGCAG
```

```
@FORJUSP02
CCGTCAATTC
+
AAAAAAAAAA
```

Q scores (as ASCII ch

Base=T, Q=':'=25

```
@HD     VN:1.0  SO
@SQ     SN:chr20
@PG     ID:TopHat
lign-edit-dist 2 -
20 /data/user446/m
HWI-ST1145:74:C101
        CCGTGTTTAAA
C     BBDCCDDCCDDDD
    AS:i:-15
HWI-ST1145:74:C101
        TGCTGGATCAT
G     DCDDDDEDDDDDD
    AS:i:-16
HWI-ST1145:74:C101DACXX:7:1204:14760:4030        16       chr20    270877   50       100M     *        0        0
        GGCTTTATTGGTAAAAAAGGAATAGCAGATTTAATCAGAAATTCCCACCTGGCCCAGCAGCACCAACCAGAAAGAAGGGAAGAAGACAGGAAAAAACCA
C     DDDDDDDDDCCDDDDDDDDDDEEEEEEEFFFEFFEGHHHHFGDJJIHJJIJIJJJIIIIGGFJJIHIIIIJJJJJJIGHHFAHGFHJHFGGHFFFDD@BB
    AS:i:-11         XM:i:2  XO:i:0  XG:i:0  MD:Z:0A85G13    NM:i:2  NH:i:1
HWI-ST1145:74:C101DACXX:7:1210:11167:8699        0        chr20    271218   50       50M4700N50M      *        0
        0        GTGGCTCTTCCACAGGAATGTTGAGGATGACATCCATGTCTGGGGTGCACTTGGGTCTCCGAAGCAGAACATCCTCAAATATGACCTCTCG
accepted_hits.sam
```

```
##fileformat=VCFv4.3
##fileDate=20090805
##source=myImputationProgramV3.1
##reference=file:///seq/references/1000GenomesPilot-NCBI36.fasta
##contig=<ID=20,length=62435964,assembly=B36,md5=f126cdf8a6e0c7f379d618ff66beb2da,species="Homo sapiens",taxonomy=x>
##phasing=partial
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 129">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FILTER=<ID=q10,Description="Quality below 10">
##FILTER=<ID=s50,Description="Less than 50% of samples have data">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=HQ,Number=2,Type=Integer,Description="Haplotype Quality">
#CHROM POS     ID        REF    ALT     QUAL FILTER INFO                              FORMAT      NA00001        NA00002        NA00003
20     14370   rs6054257 G      A       29   PASS   NS=3;DP=14;AF=0.5;DB;H2           GT:GQ:DP:HQ 0|0:48:1:51,51 1|0:48:8:51,51 1/1:43:5:.,.
20     17330   .         T      A       3    q10    NS=3;DP=11;AF=0.017               GT:GQ:DP:HQ 0|0:49:3:58,50 0|1:3:5:65,3   0/0:41:3
20     1110696 rs6040355 A      G,T     67   PASS   NS=2;DP=10;AF=0.333,0.667;AA=T;DB GT:GQ:DP:HQ 1|2:21:6:23,27 2|1:2:0:18,2   2/2:35:4
20     1230237 .         T      .       47   PASS   NS=3;DP=13;AA=T                   GT:GQ:DP:HQ 0|0:54:7:56,60 0|0:48:4:51,51 0/0:61:2
20     1234567 microsat1 GTC    G,GTCT  50   PASS   NS=3;DP=9;AA=G                    GT:GQ:DP    0/1:35:4       0/2:17:2       1/1:40:3
```

What does a VCF File Look like?

Example

VCF header

```
##fileformat=VCFv4.0
##fileDate=20100707
##source=VCFtools
##reference=NCBI36
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality (phred score)">
##FORMAT=<ID=GL,Number=3,Type=Float,Description="Likelihoods for RR,RA,AA genotypes (R=ref,A=alt)">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##ALT=<ID=DEL,Description="Deletion">
##INFO=<ID=SVTYPE,Number=1,Type=String,Description="Type of structural variant">
##INFO=<ID=END,Number=1,Type=Integer,Description="End position of the variant">
```

Body

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	SAMPLE1	SAMPLE2
1	1	.	ACG	A,AT	.	PASS	.	GT:DP	1/2:13	0/0:29
1	2	rs1	C	T,CT	.	PASS	H2;AA=T	GT:GQ	0|1:100	2/2:70
1	5	.	A	G	.	PASS	.	GT:GQ	1|0:77	1/1:95
1	100	.	T	<DEL>	.	PASS	SVTYPE=DEL;END=300	GT:GQ:DP	1/1:12:3	0/0:20

Mandatory header lines

Optional header lines (meta-data about the annotations in the VCF body)

Reference alleles (GT=0)

Alternate alleles (GT>0 is an index to the ALT column)

Deletion

SNP

Large SV

Insertion

Other event

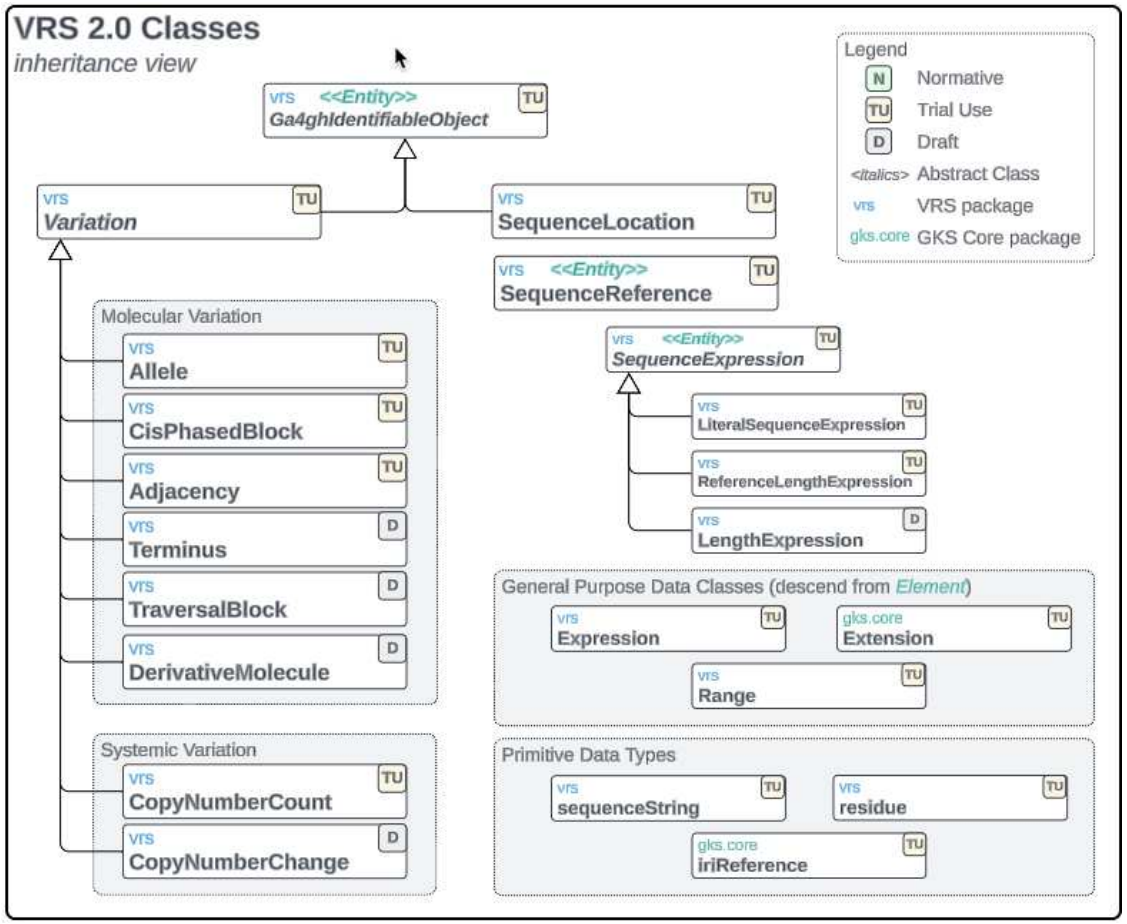
Phased data (G and C above are on the same chromosome)

https://davetang.github.io/learning_vcf_file/

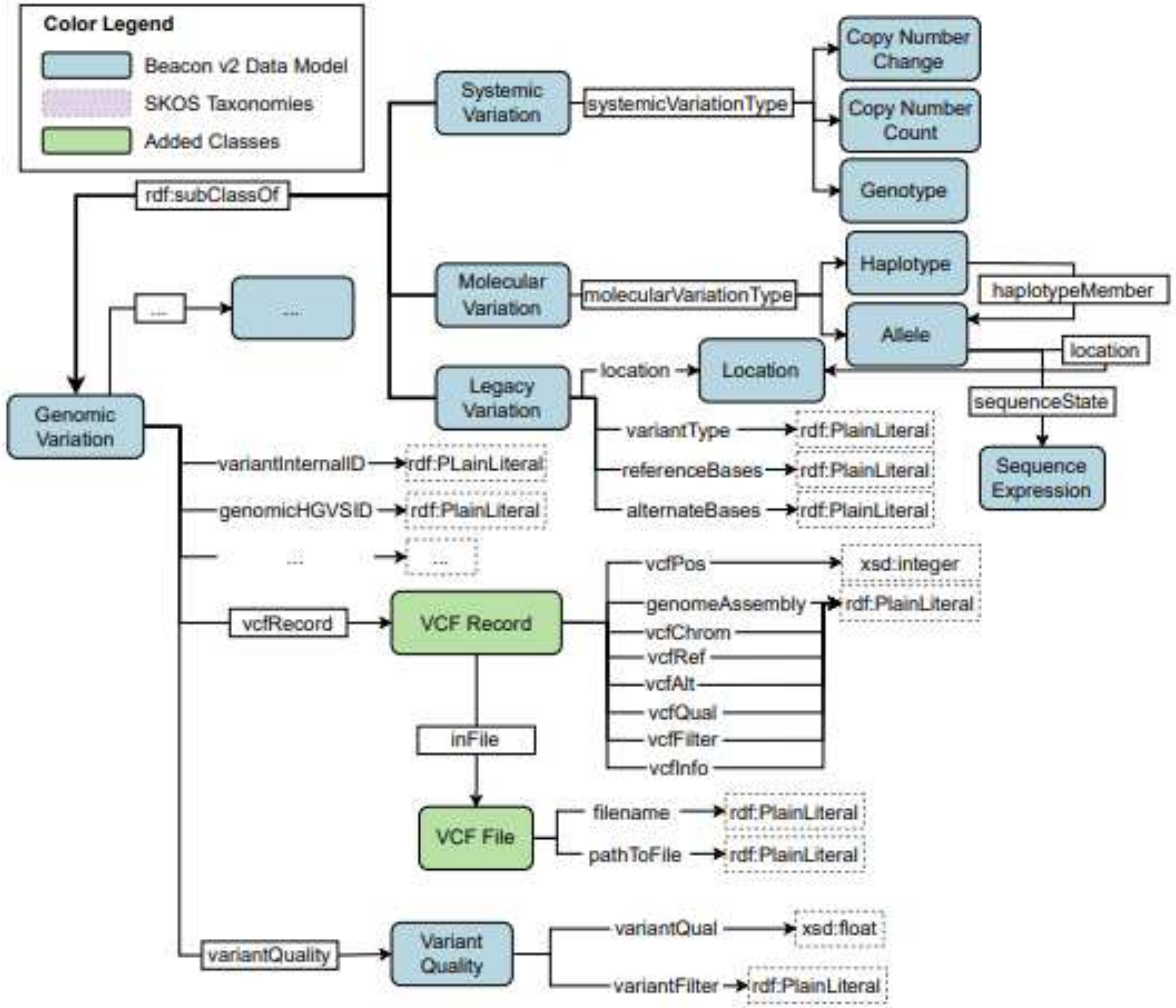
VCF Spec: https://samtools.github.io/hts-specs/VCFv4.2.pdf

Representing VCF Semantically?

GA4GH Variant Schema:



HERO Genomics Ontology:



https://vrs.ga4gh.org/en/2.0.0-ballot.2024-11/appendices/class_diagram.html#classdiagram

https://www.dei.unipd.it/~silvello/papers/2025_MCS_SWAT4HCLS.pdf

Envisioned Product Evolution

PhD End	Short Term	Medium Term	Long Term
PhD Deliverable	Start-up / Product development + Hospital implementation	Integrate with EHR + Increase application domains	Increase 3rd party application support + Scale to Europe and beyond